Name of Student: Minisha Singh	**PRN: 25070122292**
Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 26	
TITLE: Write a Java program to create and execute multiple threads using the Runnable interface.	

Problem Statement

Write a Java program to create and execute multiple threads using the Runnable interface.

Learning Objectives

To create multiple threads using the Runnable interface.

Programme

1. Create a program using Runnable to run two tasks simultaneously.

Code:

```
class Task1 implements Runnable {
    public void run() {
        for (int i = 1; i <= 5; i++) {
            System.out.println("Task 1: " + i);
        }
    }
}

class Task2 implements Runnable {
    public void run() {
        for (int i = 1; i <= 5; i++) {
            System.out.println("Task 2: " + i);
        }
    }
}

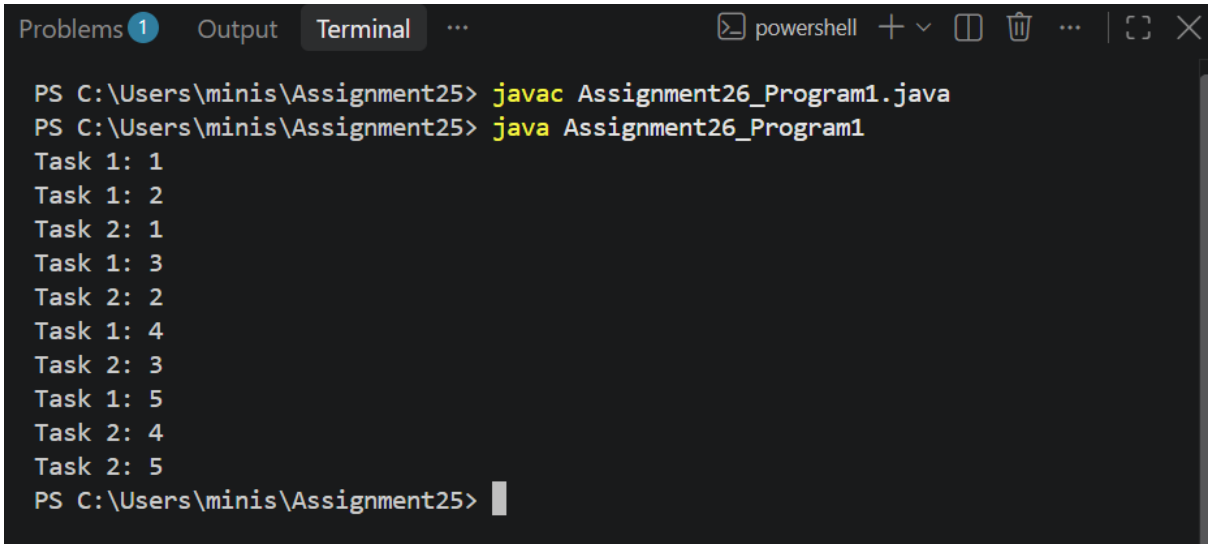
public class Assignment26_Program1 {
    public static void main(String[] args) {
        Task1 task1 = new Task1();
```

```
        Task2 task2 = new Task2();

        Thread t1 = new Thread(task1);
        Thread t2 = new Thread(task2);

        t1.start();
        t2.start();
    }
}
```

Output:



2. Create a Java program using the Runnable interface where two tasks run simultaneously: one thread performs deposit operations, and another performs withdrawal operations, both displaying results concurrently.

Code:

```
class Bank {
    int balance = 1000;

    synchronized void deposit(int amount) {
        balance += amount;
        System.out.println("Deposited: " + amount);
        System.out.println("Balance after deposit: " + balance);
    }

    synchronized void withdraw(int amount) {
        if (balance >= amount) {
            balance -= amount;
            System.out.println("Withdrawn: " + amount);
```

```
            System.out.println("Balance after withdrawal: " + balance);
        } else {
            System.out.println("Insufficient balance!");
        }
    }
}

class DepositTask implements Runnable {
    Bank bank;

    DepositTask(Bank bank) {
        this.bank = bank;
    }

    public void run() {
        bank.deposit(500);
    }
}

class WithdrawTask implements Runnable {
    Bank bank;

    WithdrawTask(Bank bank) {
        this.bank = bank;
    }

    public void run() {
        bank.withdraw(300);
    }
}

public class Assignment26_Program2 {
    public static void main(String[] args) throws InterruptedException {
        Bank bank = new Bank();

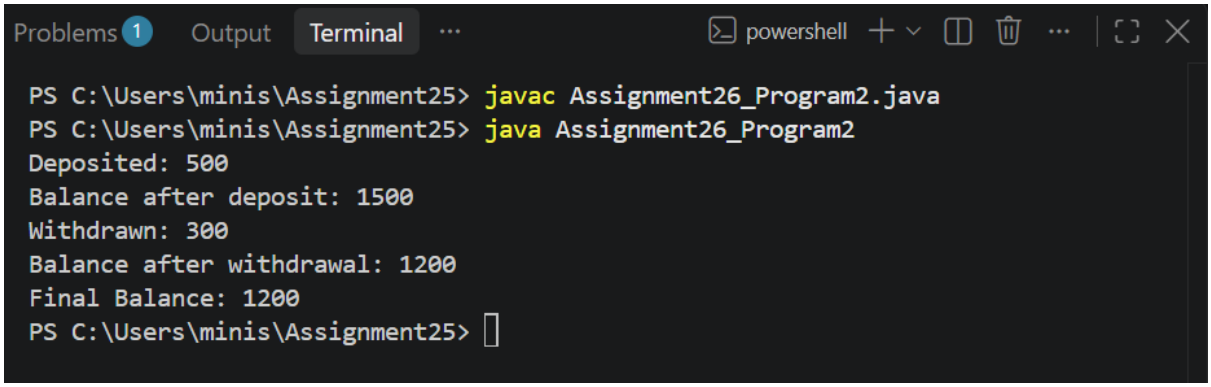
        Thread t1 = new Thread(new DepositTask(bank));
        Thread t2 = new Thread(new WithdrawTask(bank));

        t1.start();
        t2.start();

        t1.join();
        t2.join();
```

```
        System.out.println("Final Balance: " + bank.balance);
    }
}
```

Output:



Conclusion

This assignment helped us understand how to create and execute multiple threads using the Runnable interface in Java. We learned how to run two tasks simultaneously and perform bank deposit and withdrawal operations using separate threads. We also understood the importance of synchronization in managing shared resources safely. Overall, this assignment improved our understanding of multithreading, concurrent execution, and the use of the Runnable interface in Java.